

ClarityHub – Project Documentation (v1)

1. Introduction

ClarityHub is an AI-powered internal dashboard that helps users derive insights from raw CSV data. The tool allows users to upload spreadsheets, query data in natural language, and view key visualizations — all in a clean, intuitive interface.

It is built for small teams, startups, analysts, and internal operators who want instant clarity from messy datasets.

2. Goals

- Simplify data exploration using AI.
 - Empower teams to extract insights from CSVs instantly.
 - Offer a seamless dashboard experience for non-technical users.
 - Create a scalable full-stack project to showcase advanced skills.
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3. Core Features (MVP)

a. User Authentication

- JWT-based login/register system
- Role-based access: `user` , `admin`
- Password hashing with `bcrypt`

b. CSV Upload

- Drag-and-drop file upload (Multer on backend)
- File validation and error handling
- Store CSV file metadata in database

c. AI Query Engine

- Use OpenAI GPT-4 API
- Accept user’s natural language questions
- Respond with insights based on uploaded CSV

d. Dashboard View

- Display parsed CSV data
- Show summary (total rows, columns, null values, etc.)
- AI-generated KPIs and visualization-ready data

e. Upload History

- List of previously uploaded files
 - Show status (Success / Failed)
 - Timestamp, file size, and actions
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4. Tech Stack

Frontend:

- `Next.js` (App Router)
- `Tailwind CSS`
- `shadcn/ui`
- `next-themes` (theme toggling)
- `Recharts` or `Chart.js` for charts

Backend:

- NestJS (modular, scalable backend)
- Prisma ORM
- PostgreSQL DB
- Multer for file uploads
- OpenAI SDK

5. Folder Structure

Backend (/server)

```
/src
  /auth
  /user
  /upload
  /ai
  /dashboard
  /common
prisma/
```

Frontend (/app)

```
/app
  /dashboard
  /auth
  /upload
  /settings
/components
/lib
/utils
```

6. Security

- JWT for access tokens
- Backend validation using DTOs & class-validator
- Limit file size, type for upload
- Environment variables for secrets

7. Future Scope (v2 & Beyond)

- Stripe integration for subscriptions
- Multi-tenant support (multiple organizations)
- Admin panel for user management
- Real-time collaboration (e.g. shared dashboards)
- S3 bucket storage
- Notion-style workspace UI
- Advanced chart builder
- AI agent chaining (SQL → chart → explain)

8. UI Theme Strategy

- Default: **Minimal Light**
- Toggle: Light / Dark mode via next-themes
- Fonts: Inter , Manrope
- Palette: Neutral grays + Indigo/Blue highlights

9. Confidentiality

This document is part of an internal project and is not to be distributed or made public without prior authorization. All tech, designs, and ideas are original and proprietary.

We're now ready to **begin backend setup**.
Shall I write the NestJS project initialization and Auth module next?